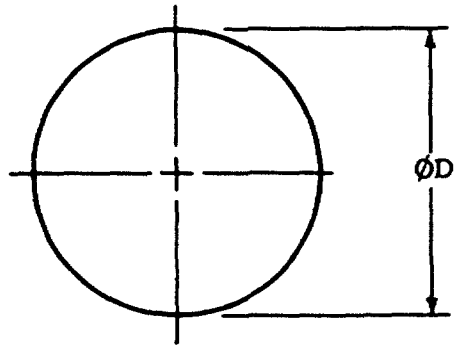
# MILITARY SPECIFICATION SHEET
# BALLS, BEARING, NICKEL-COPPER ALLOY (K-MONEL), METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bearing described herein shall consist of this specification and the latest issue of DOD-B-82737.




NOTES:

1. All dimensions are in millimeters. Column headings in tolerance tables are in accordance with AFBMA-STD-10.
2. For tolerances by grade see Table II for individual balls and Table III for lots of balls. Tolerances are in micrometers ($\mu$m).
3. Grade 10 balls are preferred in most military applications.
4. Referenced government (or nongovernment) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

FIGURE 1. BALLS, BEARING, NICKEL-COPPER ALLOY (K-MONEL), METRIC.



FSC 3110

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TABLE I. Dash numbers and dimensions.

| ØD NOM | GRADE 100 | GRADE 200 |
|---|---|---|
| | DASH NUMBERS | |
| 1.5 | 10015 | 20015 |
| 2 | 10020 | 20020 |
| 2.5 | 10025 | 20025 |
| 3 | 10030 | 20030 |
| 3.5 | 10035 | 20035 |
| 4 | 10040 | 20040 |
| 4.5 | 10045 | 20045 |
| 5 | 10050 | 20050 |
| 5.5 | 10055 | 20055 |
| 6 | 10060 | 20060 |
| 6.5 | 10065 | 20065 |
| 7 | 10070 | 20070 |
| 7.5 | 10075 | 20075 |
| 8 | 10080 | 20080 |
| 8.5 | 10085 | 20085 |
| 9 | 10090 | 20090 |
| 10 | 100100 | 200100 |
| 11 | 100110 | 200110 |
| 11.5 | 100115 | 200115 |
| 12 | 100120 | 200120 |
| 13 | 100130 | 200130 |

| ØD NOM | GRADE 100 | GRADE 200 |
|---|---|---|
| | DASH NUMBERS | |
| 14 | 100140 | 200140 |
| 15 | 100150 | 200150 |
| 16 | 100160 | 200160 |
| 17 | 100170 | 200170 |
| 18 | 100180 | 200180 |
| 19 | 100190 | 200190 |
| 20 | — | 200200 |
| 21 | — | 200210 |
| 22 | — | 200220 |
| 23 | — | 200230 |
| 24 | — | 200240 |
| 25 | — | 200250 |
| 26 | — | 200260 |
| 28 | — | 200280 |
| 30 | — | 200300 |
| 32 | — | 200320 |
| 34 | — | 200340 |
| 35 | — | 200350 |
| 36 | — | 200360 |
| 38 | — | 200380 |
| 40 | — | 200400 |
| 45 | — | 200450 |

REQUIREMENTS:

1. Material: Nickel-copper alloy (K-Monel) conforming to the chemical composition of UNS N05500 in accordance with QQ-N-286 and as specified in procurement document.

2. Hardness: Shall have a minimum hardness of 27 HRC or equivalent.

3. Surface Roughness: Tolerance limits as specified in Table II and in accordance with ANSI B46.1.

4. Material Density: Shall be 8.470 grams per cubic centimeter.

5. Part Number: Consists of the letter "D", the basic number of this specification sheet and a dash number taken from Table I.

Example: D82737/6-100120

- Dash number
- Specification sheet number

D82737/6-100120 Indicates a grade 100 bearing ball of nickel-copper alloy (K-Monel) with a basic diameter of 12 millimeters.



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TABLE II. Tolerances by grade for individual balls.

| GRADE | ALLOWABLE BALL DIAMETER VARIATION 1/ | ALLOWABLE DEVIATION FROM SPHERICAL FORM 1/ | MAXIMUM SURFACE ROUGHNESS ARITHMETICAL AVERAGE 1/ |
|---|---|---|---|
| | $V_D$ | W | |
| 100 | 2.5 | 2.5 | 0.125 |
| 200 | 5 | 5 | 0.2 |

1/ Tolerances in micrometers ( $\mu$ m)

TABLE III. Tolerances by grade for lots of balls.

| GRADE | ALLOWABLE LOT DIAMETER VARIATION 1/ | BASIC DIAMETER TOLERANCE 1/ |
|---|---|---|
| 100 | 5 | ±12.5 |
| 200 | 10 | ±25 |

1/ Tolerances in micrometers ( $\mu$ m)

Custodians:
Army - AT
Air Force - 99

Review activities:
Navy - SH
Air Force - 84
DLA - IS

Preparing activity:
Navy - OS

Agent:
DLA - IS

(Project 3110-0737-06)



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